

NAME: MEET KALAL
SIR: MR. KAMLESH SHUKLA
SUBJECT: CYBER SECURITY

Assignment

module 3: Understanding and Maintenance of Networks

Section 1: Multiple Choice

1. What is the primary function of a router in a computer network?

- (a) Assigning IP addresses to devices
- (b) Providing wireless connectivity to devices
- (c) Forwarding data packets between networks
- (d) Managing user authentication and access control

Ans: (c) Forwarding data packets between networks

2. What is the purpose of DNS (Domain Name System) in a computer network?

- (a) Encrypting data transmissions for security
- (b) Assigning IP addresses to devices dynamically
- (c) Converting domain names to IP addresses
- (d) Routing data packets between network segments

Ans: (c) Converting domain names to IP addresses

3. What type of network topology uses a centralized hub or switch to connect all devices?

- (a) Star
- (b) Bus
- (c) Ring
- (d) Mesh

Ans: (a) Star

4. Which network protocol is commonly used for securely accessing and transferring files over a network?

- (a) HTTP
- (b) FTP
- (c) SMTP
- (d) POP3

Ans: (b) FTP

Section 2: True or False

5. A firewall is a hardware or software-based security system that monitors and controls incoming and outgoing network traffic based on predetermined security rules.

Ans: True

6. DHCP (Dynamic Host Configuration Protocol) assigns static IP addresses to network devices automatically.

Ans: False

7. VLANs (Virtual Local Area Networks) enable network segmentation by dividing a single physical network into multiple logical networks.

Ans: True

Section 3: Short Answer

8. Explain the difference between a hub and a switch in a computer network.

Ans: Here's the difference between a **Hub** and a **Switch** in simple terms:

Feature	Hub	Switch
Function	Broadcasts data to all devices in the network	Sends data only to the specific device (destination)
Efficiency	Less efficient – causes unnecessary traffic	More efficient – reduces network congestion
OSI Layer	Works at Physical Layer (Layer 1)	Works at Data Link Layer (Layer 2) (sometimes

Feature	Hub	Switch
	1)	Layer 3 for advanced switches)
Intelligence	Dumb device – doesn't know where data should go	Smart device – learns MAC addresses and forwards data accordingly
Speed	Slower (because of collisions)	Faster (collision-free communication)
Security	Less secure (any device can capture data)	More secure (data goes only to intended device)

9. Describe the process of troubleshooting network connectivity issues.

Ans: Steps to Troubleshoot Network Connectivity Issues:

Check cables/Wi-Fi connection. Verify IP settings (DHCP/static).

- 1) Ping self (127.0.0.1) → Ping gateway → Ping internet IP (8.8.8.8).
- 2) Test DNS by pinging a domain (google.com).
- 3) Restart router/switch if needed.
- 4) Use `tracert`/`traceroute` for deeper issues.
- 5) Escalate to ISP/admin if unresolved.

Section 4: Practical Application

10. Demonstrate how to configure a wireless router's security settings to enhance network security.

Ans: Steps to Secure a Wireless Router:

- 1) Change default admin password.
- 2) Use **WPA3/WPA2** encryption.
- 3) Set a strong Wi-Fi password.
- 4) Change SSID (avoid default/personal info).
- 5) Disable WPS.
- 6) Enable firewall.
- 7) Update firmware.

8) (Optional) Use MAC filtering & Guest Wi-Fi.

Section 5: Essay

11. Discuss the importance of network documentation and provide examples of information that should be documented.

Ans: Importance of Network Documentation

Network documentation is **critical** for managing, troubleshooting, and securing a network.

- Helps **administrators** understand the structure and configuration.
- Speeds up **troubleshooting** when problems occur.
- Ensures smooth **network upgrades/expansions**.
- Improves **security** by tracking devices, users, and configurations.
- Acts as a **reference guide** for new IT staff or in case of emergencies.

□ **Examples of Information to Document**

1. **Network Topology Diagrams** → physical & logical layout of devices.
2. **IP Addressing Scheme** → static & dynamic IPs, subnets, gateways.
3. **Device Inventory** → routers, switches, firewalls, servers with model/serial numbers.
4. **Configuration Files** → backup of router/switch/firewall settings.
5. **User Accounts & Permissions** → who has access and at what level.
6. **Network Policies** → security rules, VLAN configurations, firewall rules.
7. **Maintenance Logs** → updates, patches, hardware replacements.
8. **ISP & Vendor Information** → contact details, service agreements.